MSG No. 64.469 *Cmca* [Type I, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,5,6]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,15,16]`

Table 3: Wyckoff site: `8c`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,13]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,16]`
5	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,9]`
6	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[6,10]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[7,11]`
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[8,12]`

Table 4: Wyckoff site: `8d`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[3,4]`
3	$[-x, 0, 0]$	`[5,6]`
4	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,8]`
5	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10]`
6	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12]`
7	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	`[13,14]`
8	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[15,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, \frac{1}{4}\right]$	`[1,3]`
2	$\left[\frac{1}{4},\, -y,\, \frac{3}{4}\right]$	`[2,4]`
3	$\left[\frac{3}{4},\, -y,\, \frac{3}{4}\right]$	`[5,7]`
4	$\left[\frac{3}{4},\, y,\, \frac{1}{4}\right]$	`[6,8]`
5	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	`[9,11]`
6	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	`[10,12]`
7	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	`[13,15]`
8	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	`[14,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `m..`

No.	position	mapping
1	$[0,\, y,\, z]$	`[1,6]`
2	$[0,\, -y,\, -z]$	`[2,5]`
3	$\left[\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	`[3,8]`
4	$\left[\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	`[4,7]`
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	`[9,14]`
6	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[10,13]`
7	$\left[0,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11,16]`
8	$\left[0,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[12,15]`

Table 7: Wyckoff site: `16g`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[x,\, -y,\, -z]$	`[2]`
3	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[-x,\, -y,\, -z]$	`[5]`
6	$[-x,\, y,\, z]$	`[6]`
7	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	`[7]`
8	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	`[9]`
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[10]`
11	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	`[14]`
15	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[16]`